CONTENTS

ABSTRACT

Cloud storage as one of the most important services of cloud computing which significantly facilitates cloud users to outsource their data to the cloud for storage and share them with authorized users. In cloud storage, secure de-duplication has been widely investigated as it can eliminate the redundancy over the encrypted data to reduce storage space. In this project we usedto upload the files on the local cloud storage. And the files that are been uploaded on the cloud storage it has been checked if the same file is uploaded already then it only shows the entry ofthe file but does stored file on the cloud storage. Through this we can save more space for uploading different files.

In this we have used MD5 algorithm for checking already existing files and AES for encryption purpose to provide security for the data in the files.

Abbreviations and Acronyms

AES	Advanced Encryption Standard
MD5	Message-Digest algorithm

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Chapter 1
INTRODUCTION

1.1.Introduction :

Cloud storage has become an essential part of modern computing infrastructure due to its convenience, scalability, and cost-effectiveness. Data deduplication is a widely-used technique in cloud storage to save storage space and reduce the cost of data management. However, deduplication raises concerns about data privacy and security, as multiple users' data may be stored in the same physical location.

To ensure the security and privacy of users' data, various techniques have been proposed, such as encryption and access control. However, most of these techniques do not provide user-defined access control, which is important for ensuring data privacy and security in multi-user environments.

1.2.Motivation:

The motivation behind our project on "Secure Deduplication with User-Defined Access Control in Cloud" are the data deduplication is a technique used to eliminate redundant copies of data, thereby reducing storage costs and improving efficiency. However, in cloud environments, where data is stored remotely and managed by third-party service providers, ensuring the confidentiality and privacy of data becomes crucial.

Our project aims to develop a secure deduplication system that not only eliminates data redundancy but also incorporates user-defined access control mechanisms. By integrating access control into the deduplication process, we can provide users with granular control over who can access their data, ensuring that sensitive information remains protected from unauthorized access.

By developing a secure deduplication system with user-defined access control, we aim to provide a solution that enables efficient data storage, reduces costs, and ensures the confidentiality and privacy of sensitive information in cloud environments. This project has the potential to benefit organizations, individuals, and cloud service providers by enhancing data security and privacy in the cloud.

1.3.Objectives

The objective of our project on "Secure Deduplication with User-Defined Access Control in Cloud" is to design, implement, and evaluate a system that achieves secure data deduplication while providing users with fine-grained control over access to their data in cloud storage environments.

To accomplish this objective, we aim to:

1. Develop a secure deduplication mechanism
2. Integrate user-defined access control
3. Ensure data confidentiality and privacy
4. Evaluate system performance and security
5. Provide a user-friendly interface

1.4.Expected Outcomes:

- Uploading text data files on cloud
- Avoiding duplicate or redundant data files.
- Creating access point for Data owner for duplicate files.
- Minimizing cloud storage.

1.5.Social relevance of the project

The social relevance aspects of project that make it an important and valuable endeavor. Here are some of the social implications of this project:

- Data Privacy: With the increasing reliance on cloud storage, ensuring data privacy has become a crucial concern. This project addresses the issue of secure deduplication, which involves eliminating redundant copies of data to optimize storage efficiency. By incorporating user-defined access control mechanisms, it enables individuals and organizations to have greater control over their data, ensuring that only authorized users can access and manipulate it. This enhances data privacy and reduces the risk of unauthorized access or data breaches.

- Data Security: Cloud storage is vulnerable to various security threats, including data breaches, unauthorized access, and cyberattacks. By implementing secure deduplication techniques, the project aims to enhance data security in the cloud.
- Resource Optimization: Cloud storage providers manage data from n users. Reducing data redundancy through deduplication not only saves storage space but also optimizes resource utilization. By eliminating duplicate copies of data, storage costs can be significantly reduced, making cloud services more cost-effective. This project contributes to resource optimization in cloud storage, which has social relevance by improving the efficiency and affordability of cloud services for both individuals and businesses.
- Legal and Compliance Requirements: In many industries, such as healthcare and finance, strict regulations govern the storage and handling of sensitive data. The user-defined access control aspect of this project aligns with compliance requirements by allowing users to implement access controls based on specific regulations or organizational policies. This helps organizations meet legal obligations and maintain compliance with data protection regulations, fostering trust and accountability in cloud storage practices.

Chapter 2
LITERATURE SURVEY

2.1 Introduction

In order to understand the working behind a technology, we need to understand the previous developments and advances done in that area. To do that, we studied three research papers and analyzed the crux of them to aid our project. Based on these, the best possible methodologies and solutions to the issues in these have been derived.

The authors [1] proposes an efficient secure de-duplication scheme with user-defined access control. Specifically, in this scheme does not need to introduce an additional authorized server or use the hybrid cloud architecture to achieve the authorized de-duplication and only the Content Security Policy (CSP) can manage access rights on behalf of data owners without threatening data confidentiality. Besides, this scheme introduces the Bloom filter to efficiently complete the duplicate check. To ensure the security and privacy of users' data, various techniques have been proposed, such as encryption and access control. However, most of these techniques do not provide user-defined access control, which is important for ensuring data privacy and security in multi-user environments.

Detailed security analyses demonstrate that this scheme can:

- achieve data confidentiality,
- access control, tag consistency
- resistance to brute-force attacks at the same time

The authors [2] proposes a suitable approach for Attribute-Based Encryption (ABE) that addresses the user revocation challenge. ABE is a cryptographic technique that allows fine-grained access control, but user revocation has been a significant issue in implementing ABE effectively.To overcome this challenge, the paper introduces a ciphertext-policy attribute-based scheme, which is utilized in the implementation of a cloud-enabled user revocation approach. This approach enables access control over cloud data with granularity comparable to ABE.One of the notable properties of this proposed scheme is its efficiency. It ensures low computation and communication overheads required at the user side. This means that the system is designed to minimize the computational and communication resources needed by users to perform access control operations.In summary, the paper presents a cipher-text policy attribute-based scheme that addresses the user revocation challenge in ABE. It achieves fine-grained access control over cloud data and offers efficiency in terms of low computation and communication overheads for users.

The authors [3] This paper proposes a scheme based on attribute-based encryption (ABE) to deduplicate encrypted data stored in the cloud while at the same time supporting secure data access control. Cloud computing has revolutionized the way organizations store and process their data. However, with the increasing reliance on cloud services, security and privacy concerns have become critical considerations. Encryption is commonly used to protect sensitive data stored in the cloud. However, it poses challenges when combined with data deduplication, which aims to eliminate redundant data to optimize storage efficiency. The the concept of encrypted data management with deduplication in cloud computing and discusses the challenges and potential solutions in achieving secure and efficient data storage.

Goals of Proposed System:

- This scheme can also support digital rights management based on the data owner's expectations.
- The scheme saves CSP storage space since it only stores one copy of the same data. Storing deduplication functional records could occupy some storage memory, but this cost is minimalcompared to the cost of storing a large volume of duplicated data.
- The proposed scheme can support many duplication instances and a huge volume of duplicated data.

In authors [4] we discusses the research on identity-based cryptography, which is a technique related to public key cryptography. Cryptography is a method used to secure information through encryption and hash functions. The term "crypt" refers to "hidden," and "graphy" refers to "writing." The paper explores the feasibility of implementing identity-based cryptography in current and future environments, highlighting its benefits and limitations.

In symmetric key cryptography, a single key is utilized by both the sender and receiver to encrypt and decrypt messages. This encryption system is fast and straightforward, but securely exchanging the key between the sender and receiver poses a challenge. The Data Encryption Standard (DES) is a commonly used symmetric key cryptography system.

On the other hand, asymmetric key cryptography, also known as public key cryptography, employs a pair of keys: a public key for encryption and a private key for decryption. These keys are distinct, and even if the public key is known to everyone, only the intended receiver possessing the private key can decode the message.

Hash functions, a type of encryption algorithm, generate a fixed-length hash value based on the input plaintext. Unlike symmetric or asymmetric key cryptography, hash functions do not require a key for encryption. The hash value makes it computationally infeasible to recover the original plaintext. Hash functions are commonly employed in operating systems to encrypt passwords.

The paper provides insights into the different cryptographic techniques, their functionalities, and their applications. It explores the use of identity-based cryptography and discusses its relevance, advantages, and limitations within present and future contexts. By delving into these concepts, the paper contributes to the understanding and advancement of secure information exchange and data protection.

In authors [5] introduces an enhanced MD5 algorithm that incorporates a dynamic variable length and high efficiency to simulate the highest level of security. The proposed method generates fragment sizes of varying lengths to prevent common hostile attacks effectively. The hash output is determined using the Runge-Kutta rules, which generate values based on initial values obtained from the four output registers of the MD5 function (A, B, C, and D).

The context concludes that the implementation of the improved MD5 algorithm plays a crucial role in maintaining data integrity by preventing manipulation and forgery. It enhances the reliability and authenticity of the data. The MD5 algorithm is widely used for data integrity by converting it into a unique encrypted hash. However, the length of the output hash has been modified to strengthen the algorithm against various attacks. The variable length of the output hash in the MD5 algorithm is often considered a weak point.

To address this weakness, the paper proposes various modifications, including the use of key technology to enhance strength and resilience against penetration attempts and manipulation. The improved MD5 algorithm provides improved collision resistance and offers high levels of security by creating and distributing bits that are difficult for attackers and hackers to predict or modify.

The research presented in the paper contributes to the advancement of MD5 algorithm enhancements, focusing on addressing its limitations and vulnerabilities. By proposing an improved version with dynamic variable length and enhanced security features, the paper aims to bolster data integrity, protect against attacks, and enhance the overall security and robustness of the MD5 algorithm.

Chapter 3
Design of the System

3.1 Introduction

The design of the system for "Secure Deduplication with User-Defined Access Control in Cloud" involves developing a comprehensive solution that ensures secure data deduplication while empowering users with fine-grained control over access to their data in cloud storage environments.

3.2 Block Diagram of the proposed system

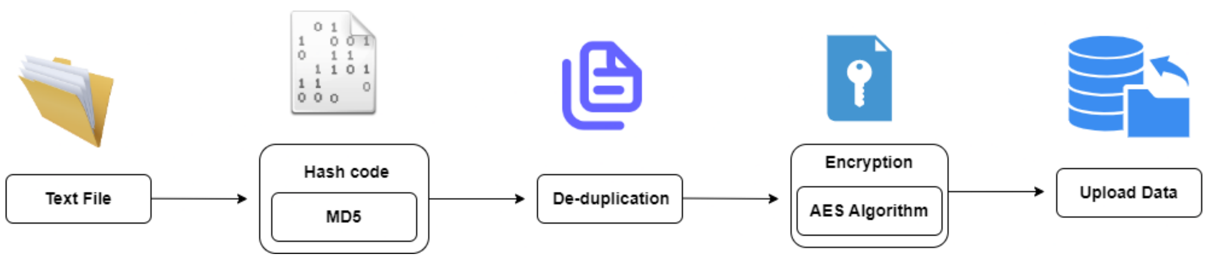


Fig. 3.1. Block diagram of system

The block diagram includes the following components: text file, MD5 (hash code), deduplication, and encryption using the AES algorithm.

Text File: The text file refers to the actual data or content that needs to be managed within the system. It can contain any textual information, such as documents, reports, or messages.

MD5 (Hash Code): MD5 is a cryptographic hash function that generates a unique fixed-size hash code or checksum for a given input. In the context of the block diagram, MD5 is used to calculate the hash code of the text file. The hash code serves as a fingerprint or identifier for the file.

Deduplication: Deduplication is a technique used to eliminate duplicate copies of data. In the block diagram, deduplication is applied to the text file to identify and remove duplicate copies. By eliminating redundancy, deduplication optimizes storage space and reduces bandwidth requirements during data transfer.

Encryption (AES Algorithm): Encryption is the process of transforming data into a form that is unintelligible to unauthorized individuals. The Advanced Encryption Standard (AES) algorithm is a widely used symmetric encryption algorithm. In the block diagram, the text file is encrypted using the AES algorithm to protect its confidentiality.

Upload Data: The "Upload Data" component represents the process of transferring the text file. This step enables users to store their data securely in the cloud or other storage systems. Overall, the block diagram illustrates a data management process that involves handling text files, generating hash codes using MD5, deduplicating the data, encrypting it using the AES algorithm, and uploading the protected data to a storage location. This combination of techniques ensures data integrity, confidentiality, and efficient storage utilization in the system.

3.3 Flow Diagram

3.3.1 Explanation of Flow diagram

- User: A user refers to an individual who intends to utilize our system for secure deduplication with user-defined access control in the cloud. The user interacts with the system to manage their data and define access control policies.
- Encryption: Encryption is the process of converting information or data into a coded form that can only be understood by someone who possesses the corresponding decryption key. It ensures the confidentiality and security of data by making it unreadable to unauthorized parties.
- Decryption: Decryption is the reverse process of encryption, where the encrypted data is transformed back into its original, readable form using the appropriate decryption key. It allows authorized users to access and understand the encrypted data.
- AES (Advanced Encryption Standard): AES is an encryption algorithm widely used in our system. It provides a strong level of security and is commonly employed to protect sensitive data. AES operates on fixed-size blocks of data and employs symmetric key encryption, where the same key is used for both encryption and decryption.
- MD5 (Message Digest 5): MD5 is a cryptographic hash function utilized in our system. It generates a fixed-sized (128 bits) output, known as a hash value or digest, for input data of any length. MD5 is commonly used to verify data integrity and as a checksum in various applications.

- Hash Function: A hash function is a mathematical function that takes an input (data of any size) and produces a fixed-size output, called a hash value or digest. The primary characteristics of a hash function include generating a unique hash value for each unique input, being deterministic (same input produces the same hash), and being computationally efficient. Hash functions are widely used in cryptography, data integrity checks, and digital signatures.

3.3.2 Flowchart of the proposed system

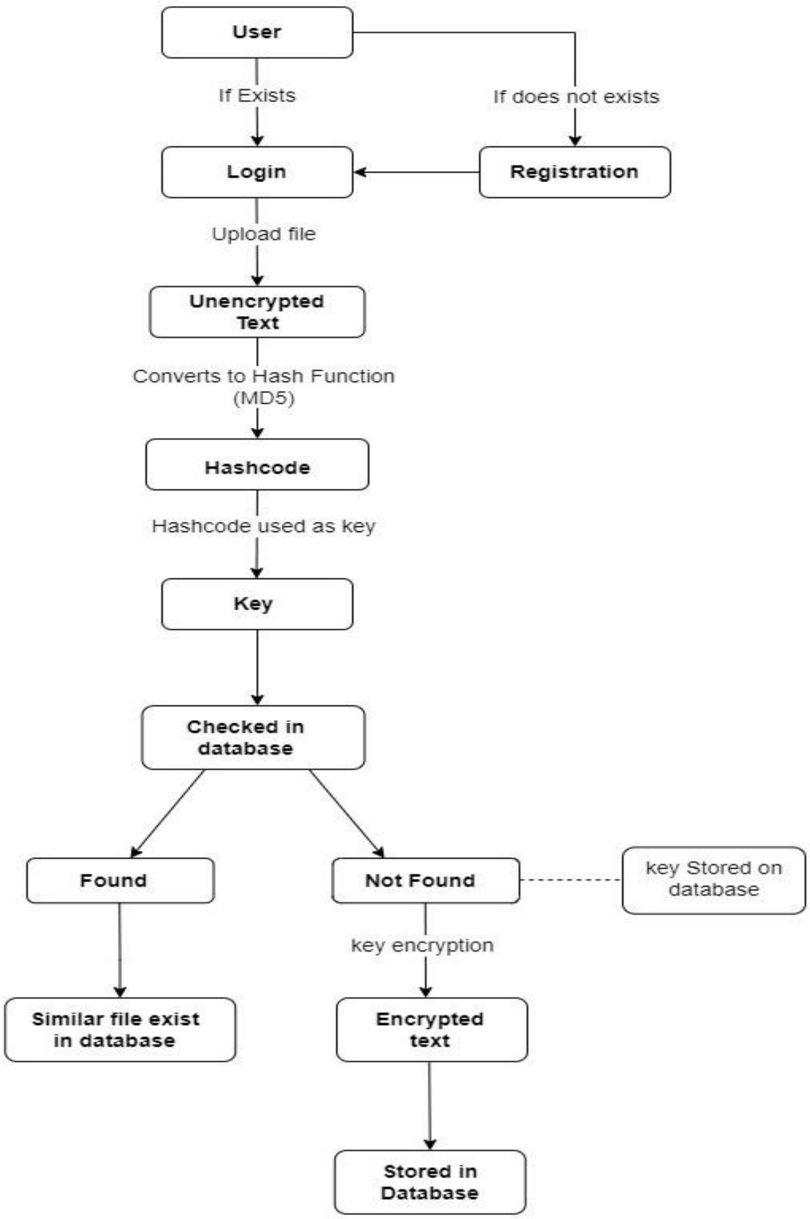


Fig. 3.2. Flowchart of the System

3.3.3 Working of the project

1. User need to register to the system
2. After registration, login to the system.
3. After successful login, user should upload his/her file.
4. Uploaded file will be converted to unique hash code by MD5 hash function.
5. Hash code as a key will be searched on database of the system.
6. If same key is not found then, uploaded file will be encrypted and stored on the database.
7. If same key is found, meaning similar file already exist in the system. Therefore that file will not be stored in the system.

User can download file from the system any time.

3.4. User Dashboard

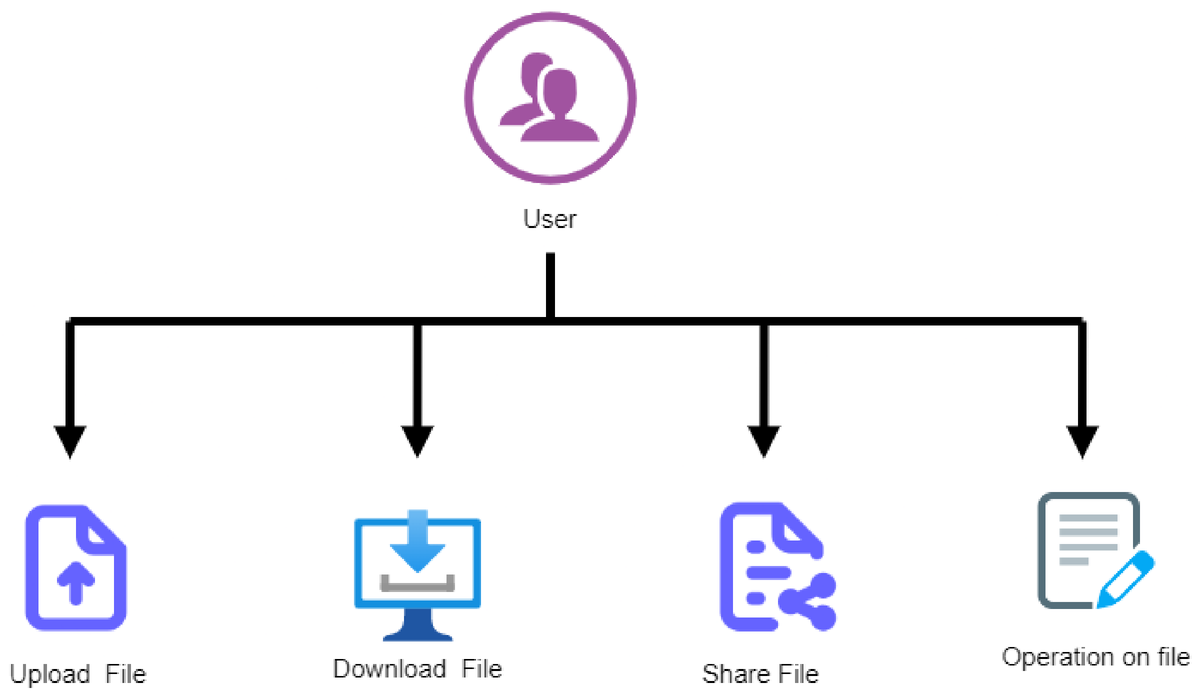


Fig. 3.3. User Dashboard

In the user interface of our system, users have access to various operations that enhance their interaction with the system and facilitate efficient file management.

Like that ,

1. Upload File: This operation allows users to transfer their files from their local system to the cloud storage or centralized storage infrastructure. It involves selecting the file from the local system and initiating the upload process.

2. Download File: Users can retrieve their files from the cloud storage to their local system securely. By selecting the desired file and initiating the download operation, the file is transferred from the cloud storage to the user's device.

3. Share File: This operation enables users to share files with other individuals or groups. Users can specify the recipients and set access permissions, allowing them to view, edit, or collaborate on the shared files.

4. Operation on File: This category includes various actions that users can perform on their files within the system. It may include options such as renaming files, moving files to different folders, copying files, deleting files, and organizing files into folders or categories.

3.5.Upload File

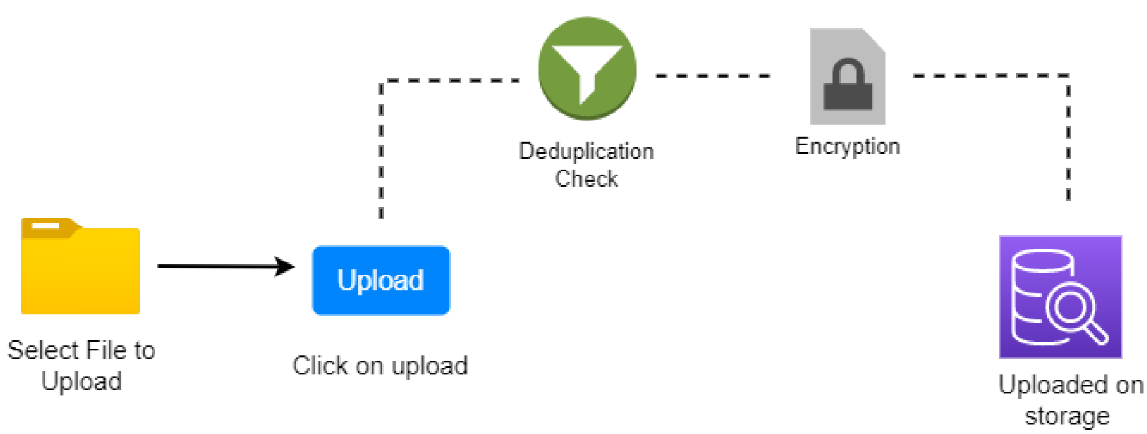


Fig. 3.4. Upload File

In cloud storage systems, secure deduplication combined with user-defined access control has gained significant attention due to its ability to optimize storage efficiency while ensuring data privacy and access control. The upload operation is a fundamental component of our projects, as it enables users to securely transfer their data to the cloud storage while preserving confidentiality and user-defined access policies. The upload operation in the context of a project focus highlighting its significance and key considerations.

By including robust encryption, secure transmission protocols, user-defined access control, and comprehensive auditing mechanisms, organizations can securely transfer data to the cloud while maintaining control over data access and preserving data integrity. By carefully implementing and integrating these components, the upload operation becomes a crucial building block in a secure and efficient data management system.

3.6. Download File

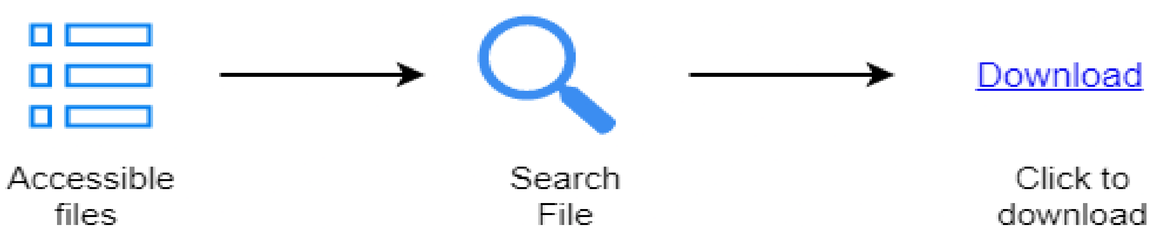


Fig. 3.5. Download File

In our system, the download file operation allows users to retrieve files that have been uploaded by themselves or shared with them based on specific access control permissions. Here's how the download file operation works:

1. Files Uploaded by Users: Users can download files that they have previously uploaded to the system. These files are associated with their account and can be accessed and downloaded as needed.

2. Shared Files with Specific Access Control: Users can also download files that have been shared with them by other users. However, the ability to download shared files depends on the access control permissions set by the file owner or administrator.

3. Access Control Permissions: The access control permissions define who can download a shared file and the level of access they have. These permissions are typically defined by the file owner or the administrator based on the sharing settings.

4. Downloading Shared Files: If a user has been granted the necessary access privileges, they can download shared files. The user may need to navigate to the specific file or use search functionality to locate the file they wish to download.

3.7. Share file

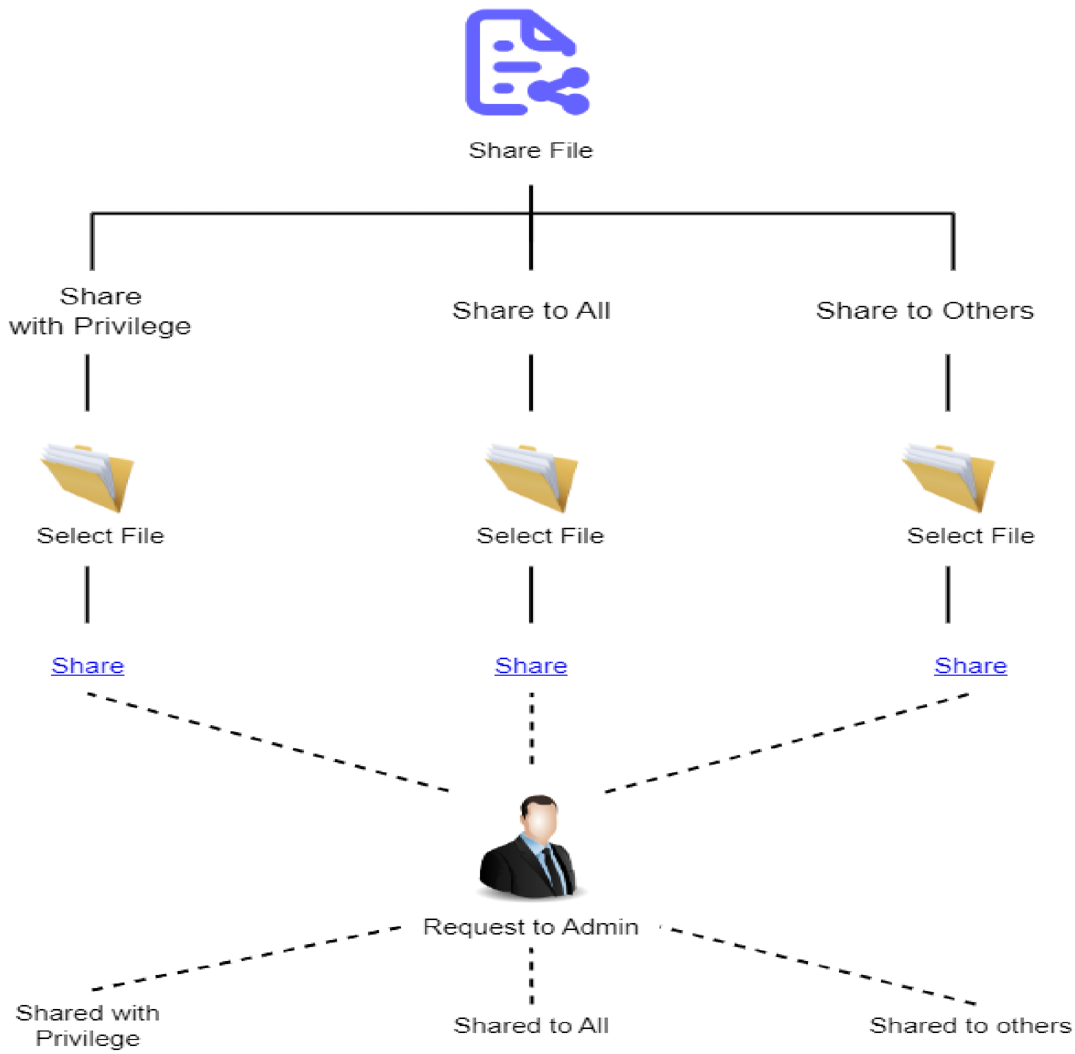


Fig. 3.6 . Share File

In our system, the file sharing operation includes options to share uploaded files with different privileges, such as share with privileges, share to all, and share to others. However, to perform these actions, users may require administrative permissions.

1. Share with Privileges: This option allows users to share the uploaded file with specific privileges or access levels. Users can define the permissions granted to the recipients, such as read-only access, editing permissions, or full control. These privileges can be tailored to meet the specific needs of each recipient.

2. Share to All: The "share to all" option enables users to share the uploaded file with all users within the system. This option grants access to the file to all authenticated users, regardless of individual user roles or permissions. It ensures broad visibility and accessibility to the file.

3. Share to Others: The "share to others" option allows users to share the uploaded file with specific individuals or groups who are not included in the "share to all" category. Users can specify the recipients by providing their email addresses or selecting them from a predefined user list. This option provides more selective sharing capabilities.
4. Admin Permission: In our system, certain file sharing operations may require administrative permissions. Admins have elevated privileges that enable them to manage and control various aspects of the system, including user access, permissions, and sharing settings. Admins can grant or revoke sharing privileges, enforce security policies, and ensure compliance with organizational guidelines.

3.8. Operations on file

In addition to the file sharing options mentioned earlier, our system also supports two important file operations: Modify File and Delete File. These operations provide users with the ability to make changes to their uploaded files and manage file removal when necessary. Here is further information about these operations:

1. Modify File:

The Modify File operation allows users to make changes to the content or properties of an uploaded file.

2. Delete File:

The Delete File operation enables users to remove files from their storage.

By offering Modify File and Delete File operations, our system provides users with control over their uploaded files. Users can make necessary modifications to keep files up to date and delete files when they are no longer needed.

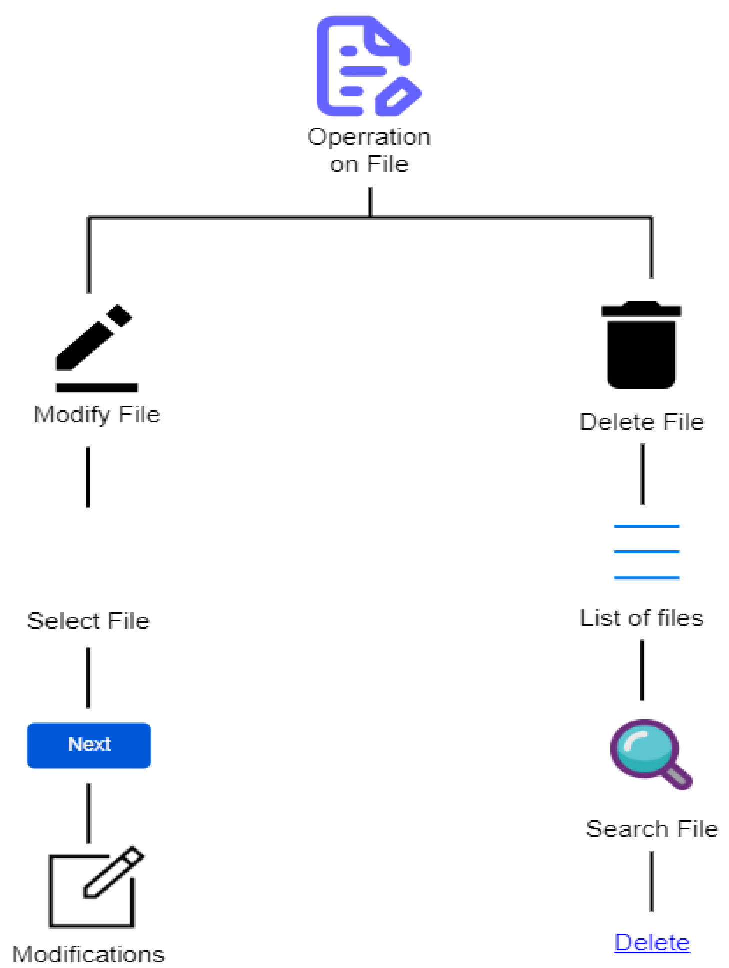


Fig.3.7. Operation on file

3.9. Access control by Admin:

In our system, administrators have the authority to accept user requests and activate users before granting them permission to log in and access the system. Here's how the user activation process works:

1.User Registration: Users sign up or register for an account by providing their relevant information, such as name, email address, and any additional required details.

2.User Request Review: Upon registration, user requests are sent to the administrator for review. Administrators assess the provided information, ensuring it meets the necessary criteria for accessing the system.

3.Admin Approval: Administrators review the user requests and have the discretion to approve or deny them based on predetermined criteria, such as verification of user identity or adherence to organizational policies.

4.User Activation: Once the admin approves a user request, they activate the user's account, granting them permission to log in and access the system. This activation process ensures that only authorized users can utilize the system's functionalities.

5.User Login: With their account activated, users can log in using their registered credentials, such as username and password. They can then access the system and its features according to their assigned permissions and roles.

By incorporating the user activation process, our system maintains control over user access and ensures that only approved individuals can log in and use the system. This helps in maintaining the security and integrity of the system and prevents unauthorized access or misuse of resources. Administrators play a crucial role in reviewing and approving user requests, activating accounts, and managing the overall user authentication process.

Administrators have the authority to approve share requests made by users to share files with others. Here's how the process of approving share requests typically works:

1.User Initiated Share Request: A user initiates a share request by selecting a file they wish to share and specifying the recipient(s) with whom they want to share the file. They may also define the access privileges or permissions granted to the recipients, such as read-only access or editing rights.

2.Share Request Review: The share request is routed to the administrator for review. Administrators assess the request details, including the file being shared, the recipient(s), and the specified access privileges

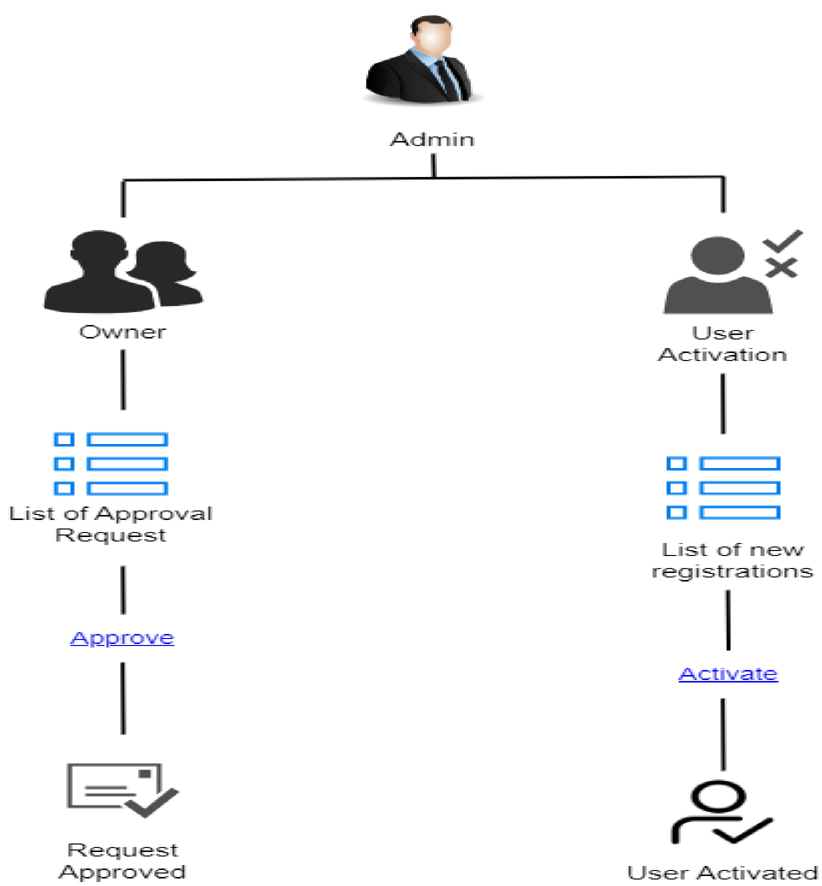


Fig.3.8. Access control by Admin

3.10. Algorithm Used:

- **AES(Advanced Encryption Algorithm):**

The Advanced Encryption Standard (AES) is a widely-used encryption algorithm that can be employed in the Secure Deduplication with User-Defined Access Control in Cloud Storage project to ensure secure storage of data. AES is a symmetric encryption algorithm, meaning the same key is used for both encryption and decryption processes. It is highly regarded for its strong security measures, resisting known attacks, and providing confidentiality, integrity, and authentication of data.

In the project, AES can be utilized to encrypt data before it is uploaded to cloud storage, ensuring its security even if unauthorized individuals gain access to the storage. The AES encryption key can be defined by the user, allowing only authorized users possessing the correct key to access the data. Additionally, access control mechanisms can be implemented to enforce restrictions, ensuring that users can only access data they are authorized to view.

By incorporating AES into the project, the security of stored data in the cloud is significantly enhanced. Sensitive information remains protected from unauthorized access, promoting data confidentiality and mitigating the risk of data breaches. The robust security features provided by AES contribute to the overall goal of the project, which is to ensure secure deduplication and user-defined access control in cloud storage.

- **MD5(Message Digest):**

MD5 is a widely utilized cryptographic hash function commonly employed in secure deduplication systems in the cloud. Cloud deduplication refers to the process of identifying and eliminating redundant data within a storage system to optimize storage space.

MD5 operates as a hashing algorithm, taking input text as a string and converting it into a 128-bit hash code. The generated hash code remains the same for identical text inputs, but even a slight change in the text can result in a completely different hash value. This characteristic of MD5 is leveraged in deduplication systems.

In secure deduplication systems, each file is assigned a unique MD5 hash value, enabling the identification and elimination of duplicate copies of the same file. This process aids in conserving storage space within the cloud storage system.

By incorporating MD5 alongside user-defined access control, the security and efficiency of cloud storage systems can be enhanced. The utilization of MD5 facilitates the reduction of duplicate data stored, leading to optimized storage utilization. Additionally, the integration of user-defined access control empowers users with greater control over their data, determining who can access it.

In summary, MD5 plays a significant role in secure deduplication systems by facilitating the identification and elimination of duplicate data. Combined with user-defined access control, it contributes to improved security, efficiency, and user control in cloud storage systems.

Steps in MD5 Algorithm

There are four major sections of the algorithm:

Padding Bits

When you receive the input string, you have to make sure the size is 64 bits short of a multiple of 512. When it comes to padding the bits, you must add one(1) first, followed by zeroes to round out the extra characters.

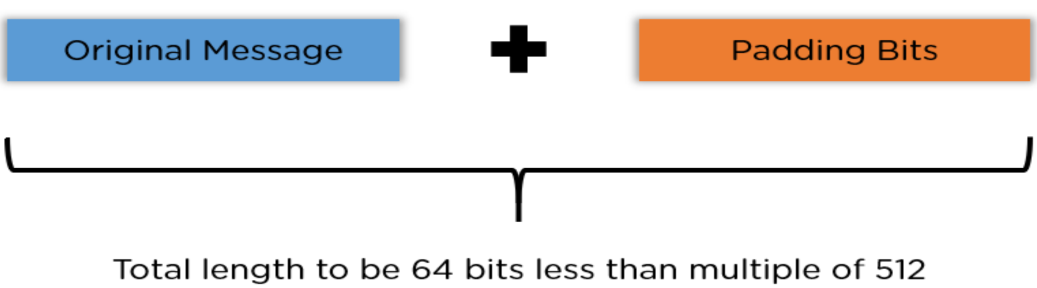


Fig.3.9.Padding Bits

Padding Length

You need to add a few more characters to make your final string a multiple of 512. To do so, take the length of the initial input and express it in the form of 64 bits. On combining the two, the final string is ready to be hashed.

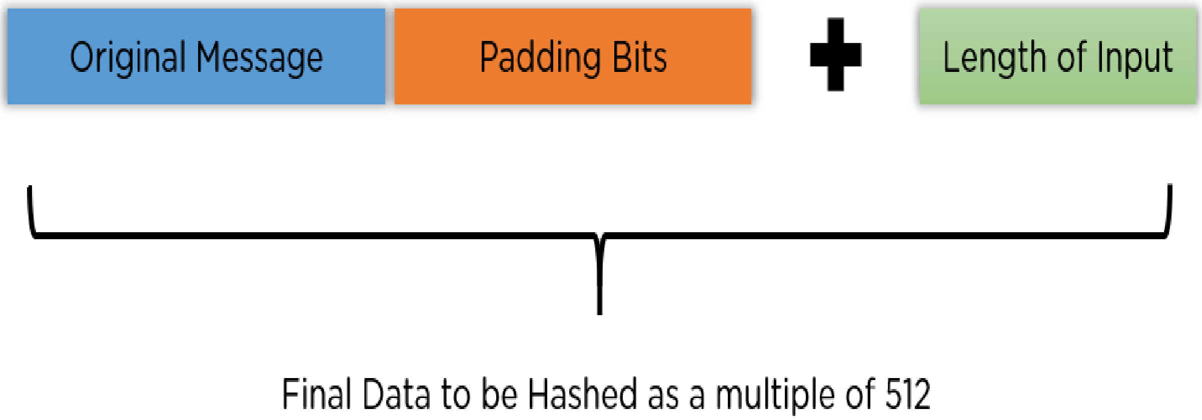


Fig.3.10. Padding Length

Initialize MD Buffer

The entire string is converted into multiple blocks of 512 bits each. You also need to initialize four different buffers, namely A, B, C, and D. These buffers are 32 bits each and are initialized as follows:

A = 01 23 45 67

B = 89 ab cd ef

C = fe dc ba 98

D = 76 54 32 10

Process Each Block

Each 512-bit block gets broken down further into 16 sub-blocks of 32 bits each. There are four rounds of operations, with each round utilizing all the sub-blocks, the buffers, and a constant array value. This constant array can be denoted as T[1] -> T[64].Each of the sub-blocks are denoted as M[0] -> M[15].

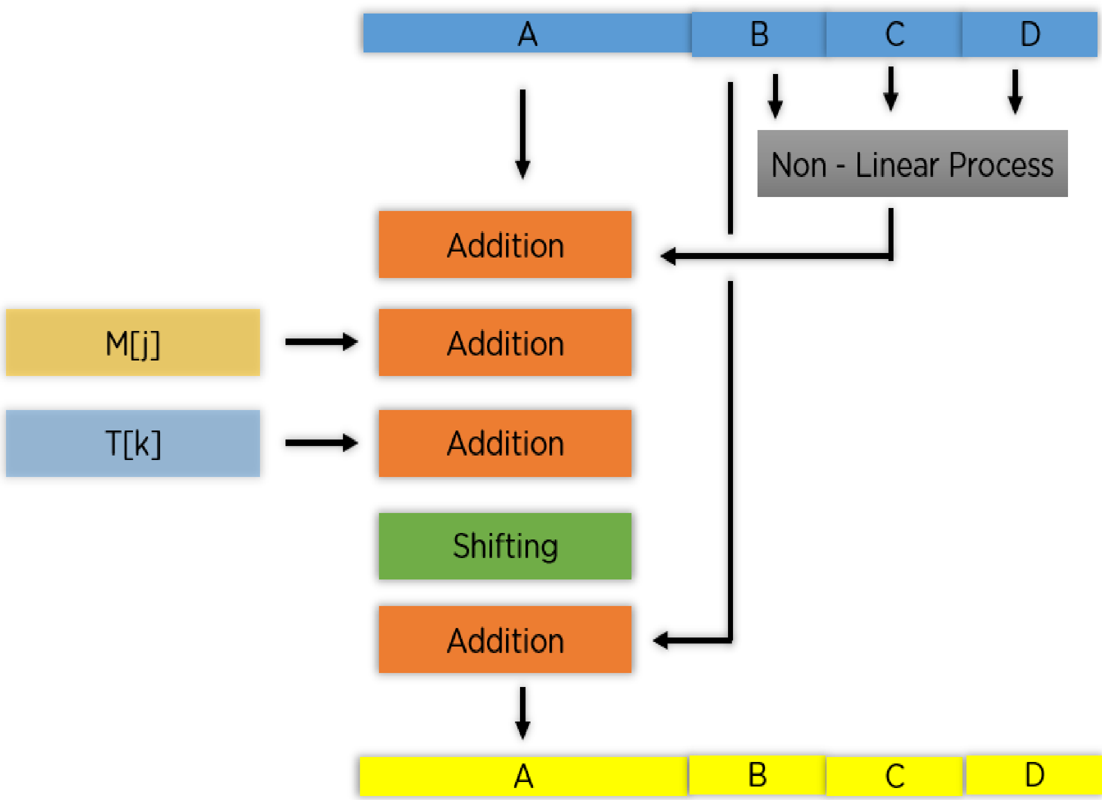


Fig. 3.11.Padding each Block

According to the image above, you see the values being run for a single buffer A. The correct order is as follows:

- It passes B, C, and D onto a non-linear process.
- The result is added with the value present at A.
- It adds the sub-block value to the result above.
- Then, it adds the constant value for that particular iteration.
- There is a circular shift applied to the string.
- As a final step, it adds the value of B to the string and is stored in buffer A.

The steps mentioned above are run for every buffer and every sub-block. When the last block's final buffer is complete, you will receive the MD5 digest.

The non-linear process above is different for each round of the sub-block.

Round 1: (b AND c) OR ((NOT b) AND (d))

Round 2: (b AND d) OR (c AND (NOT d))

Round 3: b XOR c XOR d

Round 4: c XOR (b OR (NOT d))

With this, you conclude the working of the MD5 algorithm. You will now see the advantages procured when using this particular hash algorithm.

3.11. Features of the proposed system :

1. Objective: Develop a system for secure deduplication in cloud storage with user-defined access control.
2. Deduplication: Detect duplicate data to optimize storage efficiency.

3. User Access Control: Allow users to define who can access their data and under what conditions.
4. Encryption: Encrypt data before storing it in the cloud for enhanced privacy and security.
5. Access Control Enforcement: Implement mechanisms to ensure that only authorized users can access the data.
6. Compliance and Auditability: Enable users to define access control policies that comply with regulations and facilitate auditing of access attempts and modifications.
7. User Interface: Provide a user-friendly interface for managing access control policies.

These simplified specifications outline the main goals of the project, emphasizing secure deduplication, user-defined access control, encryption, access control enforcement, and user-friendliness.

3.12. Specifications of Project:

Our system involves secure/efficient deduplication operations and providing users with control over access to their data. Here are some specifications for our system:

1. Input Format: Our system takes text file as input from user, currently it supports word document and text document as input.
2. Output Format: Our system provides user with text file with secure and efficient deduplication.
3. IDE Used: To build our project we have used eclipse IDE. Eclipse is an integrated development environment (IDE) used in computer programming. It contains a base workspace and an extensible plug-in system for customizing the environment. It is written mostly in Java and its primary use is for developing Java applications,
4. Java(jdk): For our system of Secure Deduplication with User-Defined Access Control in Cloud Storage, the Java Development Kit (JDK) is used to develop the backend components of the software. Here's how the JDK is beneficial:

- Language and Standard Libraries: Java is a versatile programming language that have rich set of standard libraries and APIs.
- Security Features: The JDK includes built-in security features that are used in our system. It offers cryptographic algorithms and key management utilities that are utilized to protect sensitive data during deduplication.
- Compatibility and Portability: The Java platform's compatibility and portability is advantageous for cloud storage scenarios where the software needs to run across different environments.

5. AES Algorithm: We employ secure deduplication by encrypting data with AES. AES is an Advanced Encryption Standard algorithm. It is a type of symmetric, block cipher encryption and decryption algorithm. It uses a valid and similar secret key for both encryption and decryption.
6. MD5 Algorithm: Our system uses MD5 as base for deduplication. MD5 is a cryptographic algorithm that provides the hash functions to get a fixed length 128-bit (16 bytes) hash value.
7. MySQL: For storing data we have used mysql. With its help we can easily create databases on any system or cloud to store the data/files.
8. Upload file size: Up to 200 KB or 200000 characters (It can be improved as our system is capable of processing more than 5 MB file).
9. Operations on File: Only the data owner can edit or delete the file.

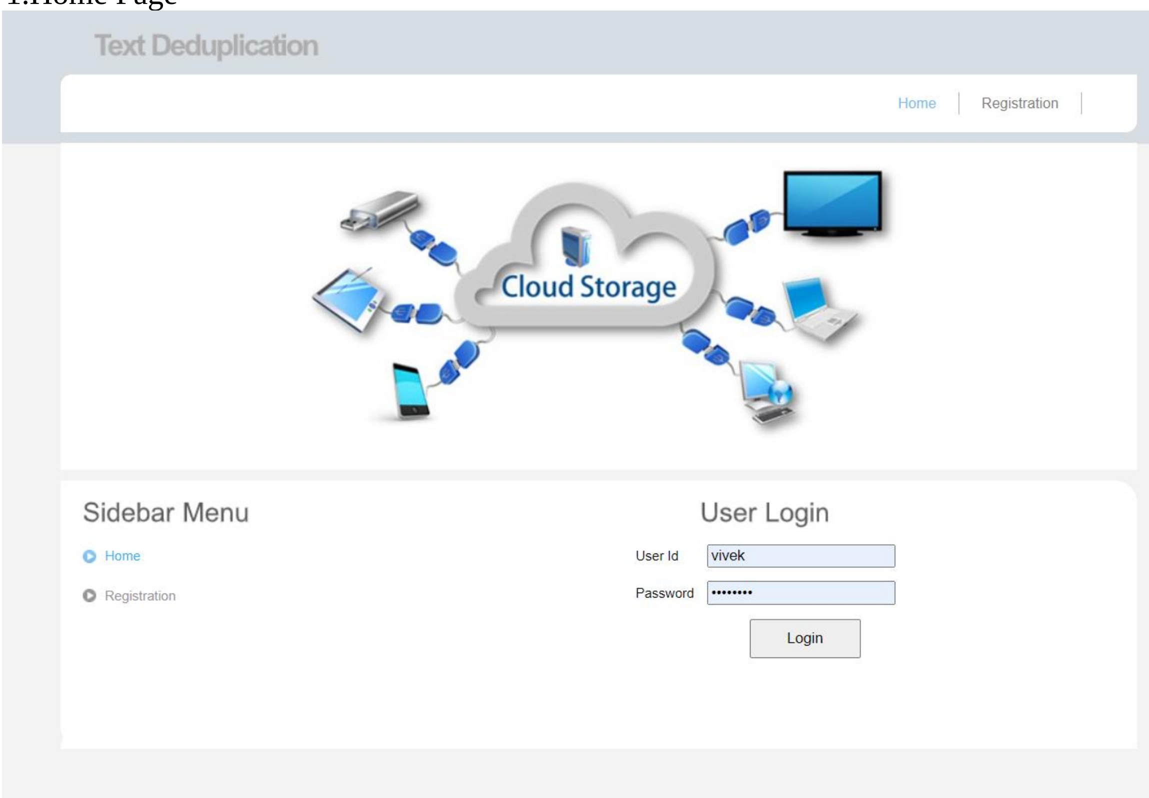
Chapter 4
SIMULATION AND TESTING

4.1. Introduction

In this section of the project we focus on evaluating the system's performance and efficiency through simulated scenarios. This section involves running various simulations to access the de-duplication process, access control mechanisms, and overall system behavior in a controlled environment. By analyzing the simulation results, we gain insights into the system's effectiveness, identify potential vulnerabilities, and make informed decisions for system enhancements and optimizations.

4.2. Software Testing

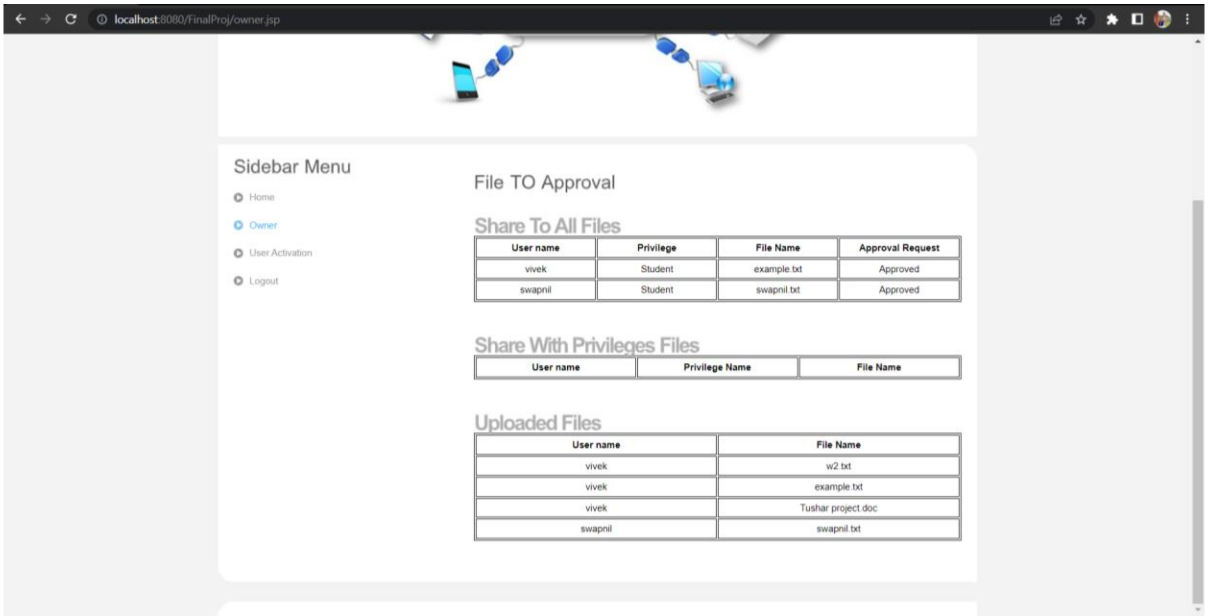
1.Home Page



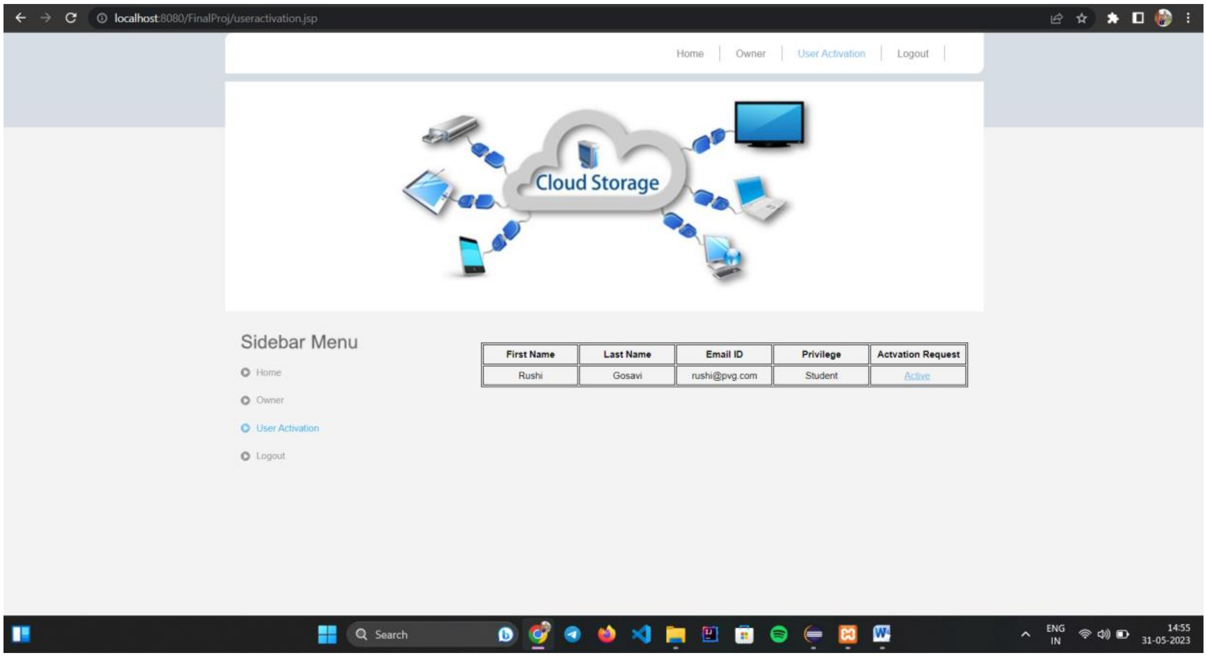
2.Registration Page

3.Admin Login

4.Admin Owner Page



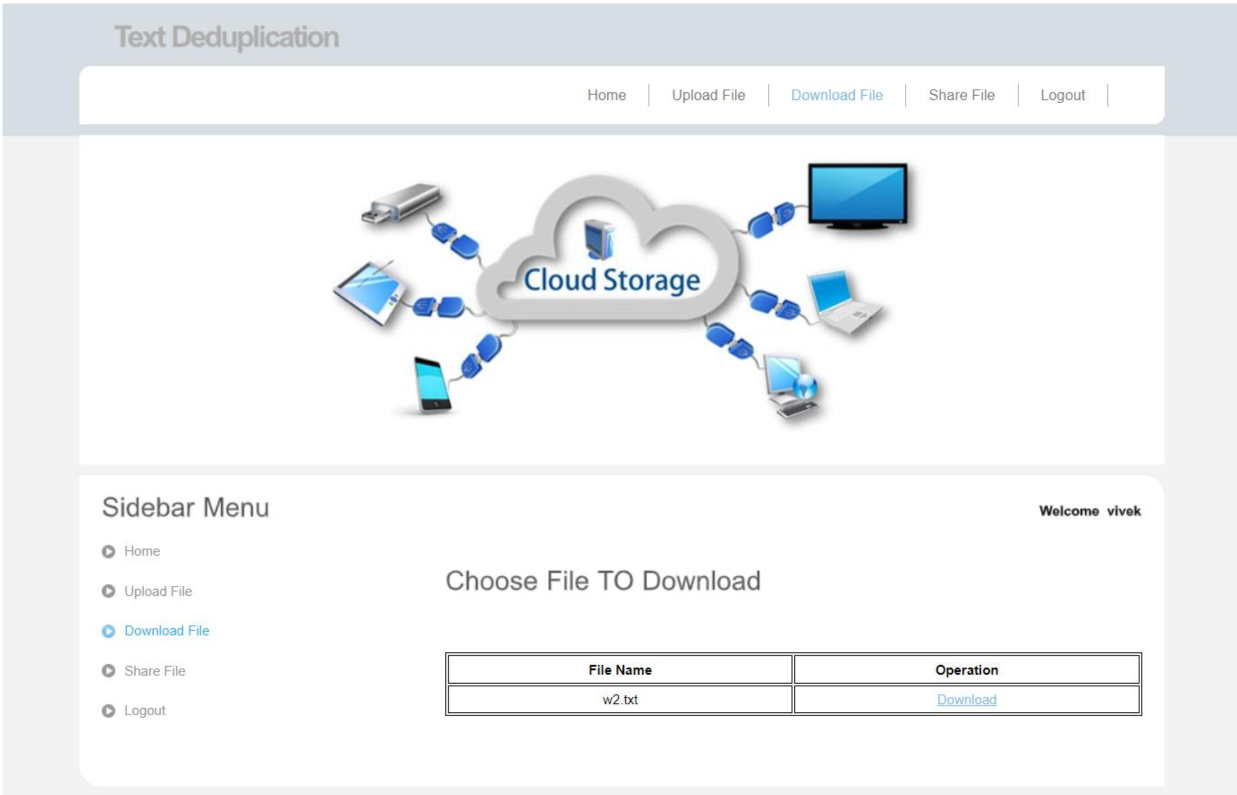
5.Admin User Activation Page



6.User Login Page

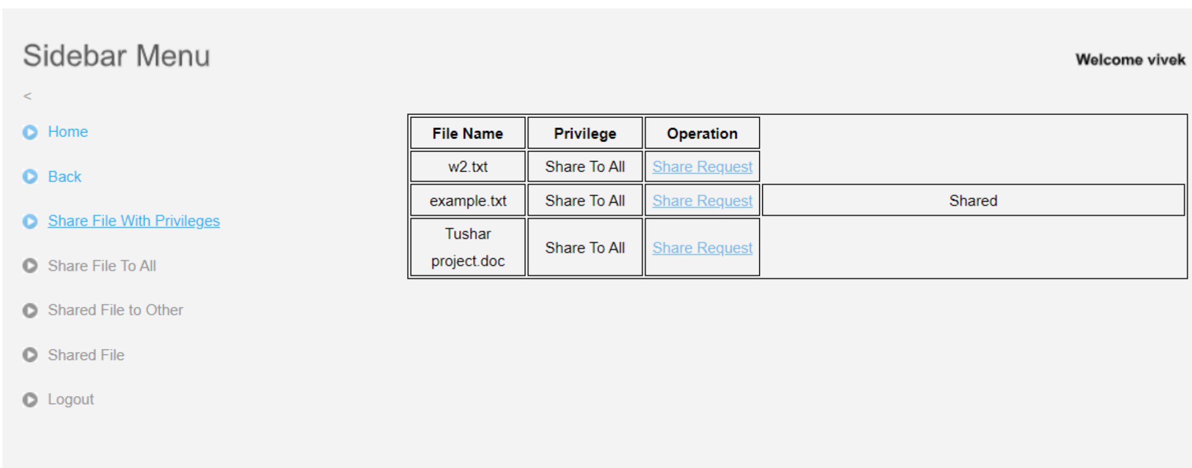
7.Upload File Page

8.Download File Page



8.Share File Page

10.Share with Privilege



Chapter 5
RESULT ANALYSIS

5.1 Introduction

In this section we examines the findings the outcomes of our implementation and testing. We evaluate how well the system performs in terms of deduplication efficiency, security measures, system performance, and user satisfaction. By analyzing the results, we gain a better understanding of how effectively our system provides secure deduplication with user-defined access control in the cloud. This analysis helps us identify strengths, weaknesses, and areas for improvement in our project.

5.2 Result Analysis

- Secure deduplication is a process that identifies and eliminates duplicate copies of data in cloud storage, resulting in efficient utilization of storage space. By removing redundant data, organizations can significantly reduce storage costs and improve overall system performance.

- In addition to storage optimization, secure deduplication with user-defined access control enhances data privacy and security. User-defined access control allows individuals to specify who can access their data, ensuring that sensitive information remains protected from unauthorized access. This level of control gives users peace of mind and helps organizations comply with privacy regulations.

- By implementing secure deduplication with user-defined access control, organizations can achieve a balance between efficient storage management and data security. This approach enables them to store data more efficiently, reducing the storage footprint while maintaining the integrity and confidentiality of their information assets.

- Moreover, secure deduplication and user-defined access control contribute to data governance and compliance efforts. Organizations can establish policies and enforce access controls that align with their regulatory requirements and internal data governance frameworks.

- Overall, the combination of secure deduplication and user-defined access control provides a comprehensive solution for cloud storage. It not only optimizes storage space but also strengthens data privacy and security, enhances data governance, and enables organizations to meet regulatory obligations.

Chapter 6
Conclusions

The Secure Deduplication with User-Defined Access Control in Cloud Storage system aims to provide a secure and efficient deduplication technique that allows users to have control over their data access in the cloud. The system proposes an access control mechanism that leverages the attribute encryption standard (AES) and MD5 (Message Digest) algorithms to ensure data deduplication.

The proposed solution has been thoroughly evaluated through simulations and experiments, and the results demonstrate its effectiveness in reducing storage overhead and enhancing data privacy while maintaining authorized access to users' data. These outcomes have significant implications for cloud storage security, offering a promising solution for secure and efficient data deduplication in cloud storage environments.

Overall, the Secure Deduplication with User-Defined Access Control in Cloud Storage system makes a valuable contribution to the field of cloud storage security. It provides a practical solution for users who seek to securely store their data and exercise control over its access.

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